$a^2+b^2=3c^2$ $\qquad a=3k_1$

$\Rightarrow (3k_1)^2+(3k_2)^2=3c^2$ $\qquad b=3k_2$

$\Rightarrow \cancel{9}(k_1^2+k_2^2)=\cancel{3}c^2$

$\Rightarrow c^2=3(k_1^2+k_2^2)$

$c^2\equiv 0 \pmod 3$

$\therefore\ 3\mid c^2 \Rightarrow 3\mid c$

$\gcd(a,b,c)=1$

but, $3\mid a$, $3\mid b$, $3\mid c$

contradiction,

$x^2+y^2-3=0$ has rational points has at least one rational points is false

so,

if, $a, b\in\mathbb{R}^{>0}$, then $a+b\geq 2\sqrt{ab}$

$(a-b)^2\geq 0$

$\Rightarrow a^2+b^2-2ab\geq 0$

$\Rightarrow (a+b)^2-4ab\geq 0$

$\Rightarrow (a+b)^2\geq 4ab$

$\rightarrow (a+b) \geq 2\sqrt{ab}$

Contradiction:

lets take $a+b < 2\sqrt{ab}$

$\rightarrow (a+b)^2 < (2\sqrt{ab})^2$

$\rightarrow a^2+2ab+b^2 < 4ab$

$\rightarrow a^2+2ab+b^2-4ab < 0$

$\rightarrow a^2-2ab+b^2 < 0$

$\rightarrow \boxed{(a-b)^2 < 0} \rightarrow$ contradiction

$\therefore$ $a+b < 2\sqrt{ab}$ is false

$a+b \geq 2\sqrt{ab}$ is true

- Suppose, $a, b \in \mathbb{Z}$, if $4 \mid (a^2+b^2)$ then a and b both are even,

Suppose,
a or b is odd or both:

Case 1: without the loss of generality,
a: odd $\quad$ b: even

$a = 2k_1+1 \quad\quad b = 2k_2$

$\therefore$ $a^2+b^2 = (2k_1+1)^2+(2k_2)^2 = 4k_1^2+4k_2^2+4k_1+1$

$= 4(k_1^2 + k_2^2 + k_1) + 1$

from modular arithmetic

$a^2 + b^2 \equiv 1 \pmod 4$

but, $4 \mid a^2 + b^2 \Rightarrow a^2 + b^2 \equiv 0 \pmod 4$

Case: 2 a & b both odd,

$a = 2k_1 + 1 \qquad b = 2k_2 + 1$

$a^2 + b^2 = (2k_1 + 1)^2 + (2k_2 + 1)^2$

$= 4k_1^2 + 4k_1 + 4k_2^2 + 4k_2 + 2$

$= 4(k_1^2 + k_1 + k_2^2 + k_2) + 2$

$a^2 + b^2 \equiv 2 \pmod 4$

$a^2 + b^2 \equiv 0 \pmod 4$